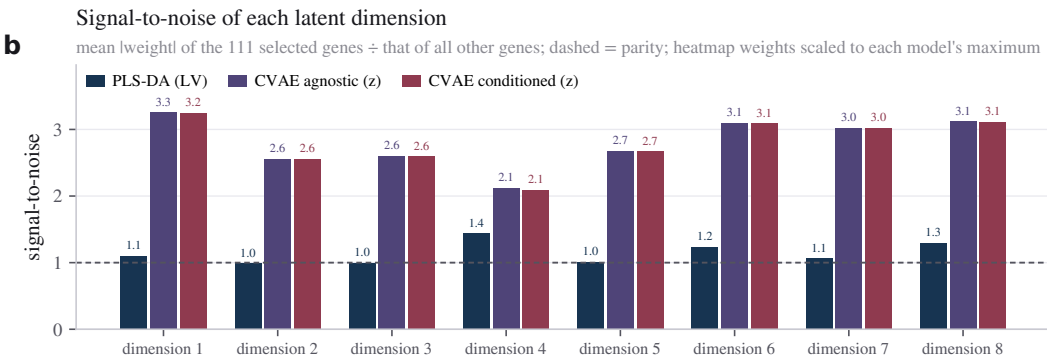
a
How each latent dimension maps back onto the 111 selected genes
PLS-DA LV
CVAE z (agnostic)
selected by
PLS-DA only
both
CVAE only
AEBP1
RHOQ
VGF
RAF1
TAB2
FOXN3
COLEC12
SPECC1L
NEDD9
SLC5A3
DUSP6
NPAS2
SLC38A2
SRSF4
ZNF148
PDE4DIP
HMGN2
TRMT5
ELMO2
ABHD8
GGA2
ELK1
TUBB2B
CCDC106
APLNR
ZMYND8
STIP1
KDM2A
RERE
FAM53C
SPRY4
COL21A1
CDK13
CPEB3
KIF5A
CAB39L
ITPKB
GFAP
RPH3A
CARTPT
BDNF
ADGRA3
ASCL1
NPTX2
NEBL
NAIP
PEG10
AKAP9
HBD
HBB
DGKG
TRIL
POLR1D
QPRT
ITPRID2
CLDN5
ZNF184
CHP1
IFITM1
ACSBG1
NINL
YEATS4
ABLIM1
KLHDC10
LAMA2
DEPP1
CDC25B
CPSF6
LRRC37A2
PSTPIP1
PLAC8
GPR17
KCTD15
ACP6
LRRC14
DPEP2
TRPM3
RCHY1
MSL2
UBE2C
LPA
DRD2
TPP2
TNR
ASF1B
STAP2
NKX2-5
HEMK1
GMIP
PWWP3A
LIPT1
CD52
ZNF669
OLR1
ANGPTL7
KLHL1
RCVRN
NAT8B
PLCXD1
TRMO
CENPQ
TOM1L1
VEGFA
GSTM1
FDXR
GREB1
FCGR2C
PPM1A
SLC2A3
TMEM80
ICAM3
PLS-DA |weight|
CVAE |weight|


b
Signal-to-noise of each latent dimension
mean |weight| of the 111 selected genes ÷ that of all other genes; dashed = parity; heatmap weights scaled to each model's maximum
PLS-DA (LV)
CVAE agnostic (z)
CVAE conditioned (z)
signal-to-noise
dimension 1
dimension 2
dimension 3
dimension 4
dimension 5
dimension 6
dimension 7
dimension 8